IT-24041

①

```
import java.util.concurrent.atomic.AtomicLong
Public class ThreadCounterLab{
    static AtomicLong safe = new AtomicLong(0)
        static long unsafe = 0;
// Thread class
static class Task extends Thread{
    int n;
        boolean mode;
          long local = 0;
  Task (int n, boolean mode){ //construct
          this.n = n;
          this.mode = mode;}

// Increment counters
    Public void run (){
      for (int i=0; i<n; i++){
        if (mode)
    safe.incrementAndGet();
    else
          unsafe ++;
          local ++;}}}
```

```
//main class
Public static void main (String [] args){
    throws InterruppledException {
  int t = Integer.parseInt (args [0]);
  int n = Integer.parseInt (args [1]);
  boolean mode = Boolean.parse Boolan
                 (args [2]);

//Reset counters
      safe.set (0);
      unsafe = 0;

Task [] th = new task[t] ;

//create and start threads
    for (int i = 0; i<t ; i++){
       th.[i] = new task (n, mode);
    th[i] .start ();}
    long total 0 ;
```

```
//wait for threads and sum local
//counters
        for (task.x : th){
            x.join();
            total += x.local;}
//calculate results.
        long s = mode ?
            safe.get(): unsafe;
    long diff = Math.abs(s-total);
double percent = total==0?0:
            diff * 100.0/total;
//Display output
System.out.println("static:" + s);
System.out.println("Non-static:"+total);
System.out.println("Difference:"+diff);
System.out.println("Percentage:%2f\n",
            percent);}}
```